

Lesson 04: Linear First-Order Differential Equations

Professor Dave Explains Differential Equations Master Series · Lesson 4

Video: Linear First-Order Differential Equations Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that when an equation cannot be separated because y and dy/dx are locked together in a sum, we use the magic of the Integrating Factor. By multiplying the entire equation by an engineered factor $\mu(x) = e^{\int P(x)dx}$, the left side magically collapses into the derivative of a single product $\frac{d}{dx}[\mu(x)y]$ via the reverse Product Rule. One simple integration then unlocks the solution!

2. Core Theoretical Framework & Rigorous Formulations

The 5-Step Integrating Factor Recipe

1. **Standard Form:** Divide by the leading coefficient so that the coefficient of y' is exactly 1:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

2. **Compute Integrating Factor:** $\mu(x) = \exp\left(\int P(x)dx\right)$ (do not add $+C$ here).

3. **Multiply and Collapse:** Multiply across by $\mu(x)$:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

4. **Integrate Both Sides:** $\mu(x)y = \int \mu(x)Q(x)dx + C$ (add $+C$ right here!).

5. **Isolate $y(x)$:** $y(x) = \frac{1}{\mu(x)} \int \mu(x)Q(x)dx + \frac{C}{\mu(x)}$.

3. Fully Solved Whiteboard Problems

Exam Master Problem: First-Order Linear ODE with Discontinuous Interval

Problem Statement: Solve the initial value problem $x\frac{dy}{dx} + 2y = 4x^2$, with $y(1) = 3$ for $x > 0$. Identify transient vs steady-state terms.

Step 1: Normalize to Standard Form

Divide by x : $y' + \frac{2}{x}y = 4x$. Here $P(x) = \frac{2}{x}$ and $Q(x) = 4x$.

Step 2: Determine Integrating Factor

$\mu(x) = e^{\int \frac{2}{x}dx} = e^{2\ln x} = e^{\ln(x^2)} = x^2$ (valid for $x > 0$).

Step 3: Multiply and Collapse

$x^2y' + 2xy = 4x^3 \implies \frac{d}{dx}[x^2y] = 4x^3$.

Step 4: Integrate Both Sides

$x^2y = \int 4x^3dx + C = x^4 + C$.

Step 5: Solve for $y(x)$ and Apply IVP

$y(x) = x^2 + \frac{C}{x^2}$. Apply $y(1) = 3 \implies 1^2 + \frac{C}{1^2} = 3 \implies C = 2$. Final solution: $y(x) = x^2 + \frac{2}{x^2}$. Transient term: $2/x^2 \rightarrow 0$ as $x \rightarrow \infty$; steady-state response: x^2 .

⚠ Common Exam Pitfall & Trapdoor Avoidance

Forgetting to divide by the leading coefficient $a_1(x)$ before finding $P(x)$ is the single most common student error in ENGR 213 exams!

4. Key Takeaways & Exam Strategy Formula Card

🌀 Core Takeaway & Exam Rule

Always normalize to $y' + P(x)y = Q(x)$ before calculating $\mu(x) = e^{\int Pdx}$.

- Video reference: [Linear First-Order Differential Equations](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.